

Project Proposal Report

Cover Page

Course Name: Advanced Database Management Systems

Project Topic Title: HCMIU Map with multiple contributors and live updates

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GitHub: <https://github.com/huynhtrankhanh/hcmiu-map-collaborative>

Abstract

Navigating and sharing knowledge within a university campus is often disjointed. This project provides an interactive, community-powered map of the Ho Chi Minh City International University campus. It combines classic pathfinding features, such as shortest route calculations and traveling-salesman-style planners, with a collaborative knowledge base. Every room, user, and comment acts as an "entity" that can be discussed, referenced, and followed. To support this highly interconnected architecture and dispute resolution features, the project will utilize **ArangoDB** as its core database.

Selected Database Management System

Selected DBMS: ArangoDB

Justification for Selection:

The project relies heavily on two distinct data paradigms: storing complex, nested documents (such as user profiles, map entities, discussions, and court trials) and mapping deep relationships between them. ArangoDB was chosen because it is a native multi-model database capable of acting as both a document store and a graph database. This allows the backend to handle everything natively without managing multiple distinct database systems.

Relevant Key Features of the DBMS:

1. **Document & Graph Storage:** ArangoDB handles standard document collections (users, entities, trials, notifications) alongside edge collections (entity_references).
2. **ArangoDB Query Language (AQL):** Used heavily for complex data retrieval, full-text search across entities, and identifying references.
3. **Graph Traversal (Degree of Separation):** The database's native graph routing allows the application to find the shortest reference distance path between entities, surfacing detailed relationships efficiently.

Project Plan and Timeline

Week / Dates	Phase	Expected Outcomes
Week 1 <i>(Feb 16 - Feb 22)</i>	Design & Orchestration	• Setup Docker Compose to orchestrate ArangoDB, the Node backend, and Vite frontend. • Define the database schema (Document and Edge collections).
Week 2 <i>(Feb 23 - Mar 1)</i>	Backend API Implementation	• Connect Express backend with the arangojs driver. • Implement the authentication flow, REST endpoints, and WebSocket real-time event distribution. • Submit Proposal Report (Due Feb 28).
Week 3 <i>(Mar 2 - Mar 8)</i>	Collaboration & Frontend Integration	• Build the collaborative layer to support tracking entity references, full-text searches, and the "Court Trial" dispute functionality. • Connect the TypeScript/Vite frontend UI.
Week 4 <i>(Mar 9 - Mar 13)</i>	Testing & Reporting	• Execute backend API tests using Node's native test runner and end-to-end tests using Puppeteer. • Finalize application documentation and write Final Report (Due Mar 13).

Tools and Programming Languages

To build, test, and containerize the application, the following technology stack is defined in the repository:

- **Selected DBMS:** ArangoDB (Multi-model Graph and Document Database)
- **Programming Languages:** TypeScript, JavaScript, HTML, CSS
- **Frameworks and Development Tools:**
 - **Frontend:** Vite
 - **Backend Server:** Node.js
 - **API Framework:** Express
 - **WebSockets:** ws module
 - **Database Driver:** arangojs
 - **Testing:** Node.js native test runner and Puppeteer (for end-to-end tests)
 - **Containerization:** Docker & Docker Compose